

ECO462 PSET HW8

Written Solutions

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Fall 2025

Last updated: December 12, 2025

Question 1: CDS-Implied Italian Sovereign Default Intensities

Conventions: quarterly payments ($\Delta = 1/4$), premium is paid even in the default quarter, protection $(1 - R)$ is paid at the end of the quarter of default, $R = 40\%$. Discounting uses continuously compounded spot rates; for intermediate maturities we linearly interpolate spot rates between the provided 1, 3, 5, 10 year points and use $D(T) = e^{-r(T)T}$. Numerical results are computed from `data_HW8.csv` in the accompanying notebook.

1(a)

Let τ be the default time and let the (annual) risk-neutral default intensity be λ , so the default probability in any quarter is $\lambda/4$. Define $t_k = k/4$ and survival $S_k = \mathbb{P}(\tau > t_k)$. Then

$$\mathbb{P}(\tau \leq t_k) = 1 - S_k, \quad S_k = (1 - \lambda/4)^k.$$

Since the buyer pays the premium at the end of quarter k whenever it has survived to the start of that quarter, the expected present value (EPV) of premium payments on a 1-year CDS with annualized spread S is

$$\text{PV}_{\text{prem}}(S; \lambda) = \sum_{k=1}^4 e^{-r(t_k)t_k} \left(\frac{S}{4} \right) S_{k-1}.$$

1(b)

The probability that default occurs exactly in quarter k is

$$\mathbb{P}(t_{k-1} < \tau \leq t_k) = S_{k-1} (\lambda/4).$$

Thus the EPV of the protection leg is

$$\text{PV}_{\text{prot}}(\lambda) = \sum_{k=1}^4 e^{-r(t_k)t_k} (1 - R) S_{k-1} (\lambda/4).$$

1(c)

The fair spread S solves $\text{PV}_{\text{prem}}(S; \lambda) = \text{PV}_{\text{prot}}(\lambda)$. Using

$$\text{PV}_{\text{prem}}(S; \lambda) = \sum_{k=1}^4 e^{-r(t_k)t_k} \left(\frac{S}{4} \right) S_{k-1}, \quad \text{PV}_{\text{prot}}(\lambda) = \sum_{k=1}^4 e^{-r(t_k)t_k} (1 - R) S_{k-1} \left(\frac{\lambda}{4} \right),$$

we obtain

$$\sum_{k=1}^4 e^{-r(t_k)t_k} S_{k-1} \frac{S}{4} = \sum_{k=1}^4 e^{-r(t_k)t_k} S_{k-1} (1 - R) \frac{\lambda}{4}.$$

Canceling the common factor $\sum_{k=1}^4 e^{-r(t_k)t_k} S_{k-1}$ yields the closed form

$$S = (1 - R)\lambda.$$

Hence, under the 1-year constant-intensity assumptions, S does *not* depend on the discount rates $\{r(t_k)\}$, and the implied intensity is

$$\lambda = \frac{S}{1 - R}.$$

Example (first month in sample). Using the July 2010 observation,

$$\lambda_{1,2010-07} = 0.017995.$$

1(d)

For a 3-year CDS, assume piecewise-constant intensities: λ_1 on $[0, 1)$ and λ_3 on $[1, 3)$. For quarter k , let $\lambda_k = \lambda_1$ if $t_{k-1} < 1$ and $\lambda_k = \lambda_3$ otherwise. Then survival evolves as

$$S_k = S_{k-1} (1 - \lambda_k/4), \quad S_0 = 1.$$

The legs generalize to

$$PV_{\text{prem}} = \sum_{k=1}^{12} e^{-r(t_k)t_k} \left(\frac{S}{4}\right) S_{k-1}, \quad PV_{\text{prot}} = \sum_{k=1}^{12} e^{-r(t_k)t_k} (1 - R) S_{k-1} (\lambda_k/4).$$

Given the observed 1-year spread, we first solve for λ_1 ; then given the 3-year spread we solve for λ_3 . For July 2010:

$$\lambda_{3,2010-07} = 0.021785.$$

1(e)

Assume $\lambda_3 < \lambda_1$ and a constant discount rate r (so $D(t_k) = e^{-rt_k}$). When r increases, later cash flows are discounted more heavily. With $\lambda_3 < \lambda_1$, default risk (and hence protection payments) is relatively front-loaded: a larger fraction of the protection leg occurs earlier than the premium leg, which continues (in expectation) into later quarters when the issuer survives. Increasing r therefore reduces the present value of the (later-weighted) premium leg more than the protection leg, so the par spread S must *increase* to restore $PV_{\text{prem}} = PV_{\text{prot}}$.

1(f)

With available 1-, 3-, 5-, and 10-year CDS spreads each month, we assume piecewise-constant intensities $\lambda_{1,t}$ on $[0, 1)$, $\lambda_{3,t}$ on $[1, 3)$, $\lambda_{5,t}$ on $[3, 5)$, and $\lambda_{10,t}$ on $[5, 10]$. For each month we bootstrap sequentially: solve $\lambda_{1,t}$ from the 1Y spread, then $\lambda_{3,t}$ from the 3Y spread given $\lambda_{1,t}$, then $\lambda_{5,t}$ from 5Y given prior, then $\lambda_{10,t}$ from 10Y.

Summary statistics (2010-07 to 2025-12). The table reports the distribution of the monthly implied intensities:

	$\lambda_{1,t}$	$\lambda_{3,t}$	$\lambda_{5,t}$	$\lambda_{10,t}$
mean	0.012900	0.025768	0.035653	0.041790
std	0.014952	0.018891	0.017413	0.011668
min	0.000500	0.003777	0.008243	0.016646
255075max	0.086352	0.101671	0.096597	0.085058

Time-series plot. Figure 1.0.1 plots the four series together.

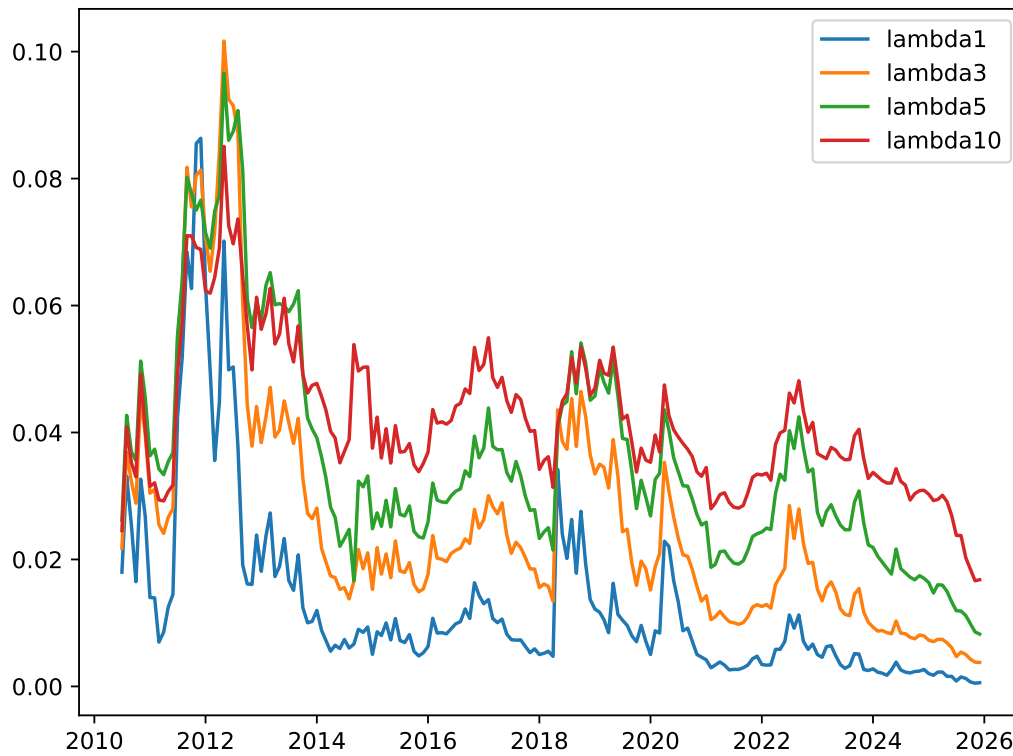


Figure 1.0.1: Monthly CDS-implied risk-neutral default intensities for Italy.

Comment on term structure. Over most of the sample, the term structure is upward sloping: $\lambda_{1,t} < \lambda_{3,t} < \lambda_{5,t} < \lambda_{10,t}$. In the data, this monotone-increasing ordering holds about 88.9% of months outside 2011–2012; the main alternative pattern is a mild hump where $\lambda_{5,t} > \lambda_{10,t}$ (about 10.5% of those months).

1(g)

In “normal” times (outside crisis periods), the typical ordering is $\lambda_{1,t} < \lambda_{3,t} < \lambda_{5,t} < \lambda_{10,t}$ (often with at most a mild hump at 5Y).

Eurozone crisis inversion check (2011–2012). The months in 2011–2012 for which $\lambda_{1,t} > \lambda_{5,t}$ are:

Month	$\lambda_{1,t}$	$\lambda_{3,t}$	$\lambda_{5,t}$	$\lambda_{10,t}$
2011-11	0.085527	0.080517	0.075020	0.069134
2011-12	0.086352	0.081347	0.076603	0.068851

An inversion $\lambda_{1,t} > \lambda_{5,t}$ indicates that investors price especially high *near-term* default risk relative to medium-term risk, consistent with acute rollover/liquidity concerns (e.g., fear of a near-term funding crisis) rather than a belief that the sovereign is fundamentally insolvent over the medium run. If CDS spreads spike but the ordering of λ is preserved (still upward sloping), that pattern suggests investors are repricing default risk upward at all horizons but still view default risk as increasing with maturity; in the Italian context, this is consistent with elevated uncertainty but continued belief that

fiscal adjustment/ECB backstops make the longer-run outlook less “knife-edge” than the short-run refinancing window.